

Satisfiability Modulo Theories

Verifying cyberphysical systems

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Some of the slides for this lecture are adapted from slides by Clark Barrett

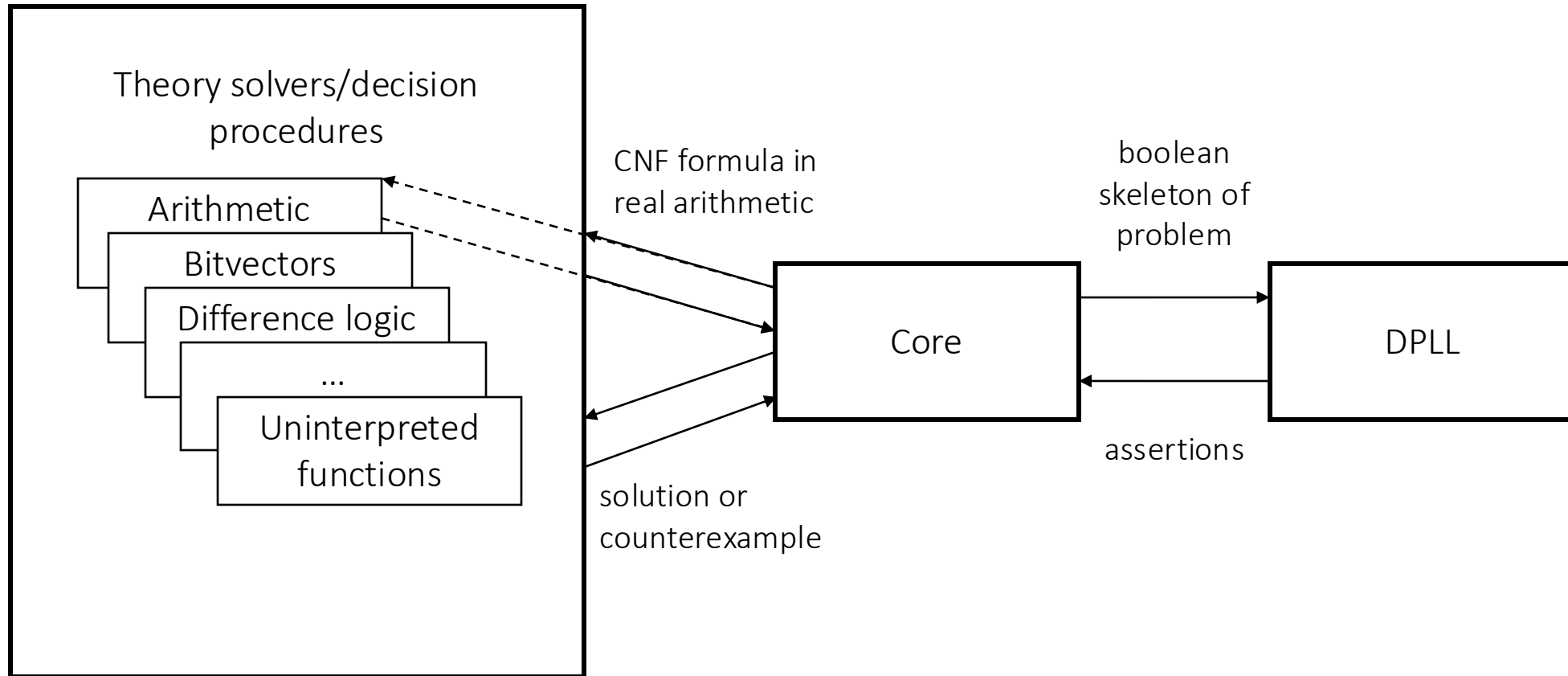
Satisfiability modulo theories

- SAT: Given a *well-formed formula* in propositional logic, determine whether there exists a satisfying solution
- A *satisfiability modulo theory (SMT)* problem is a generalization of SAT in which some of the binary variables are replaced by predicates over a suitable set of non-binary variables
- $\phi_1(w, x, y, z) := (x - y = 5) \wedge (z - y \geq 2) \wedge (z - x > 2) \wedge (w - x = 2)$
- $\phi_2(x, y, z) := (3x^2 - 4y + 5z \leq 5) \wedge (-2x + 5z^3 \leq 7)$
- ϕ_1 is a predicate in *difference logic* in which the variables are real-valued, and the clauses are constructed with standard comparison operations $>$, $\geq$, $=$ and $-$ (minus)
- ϕ_2 is a predicate in real arithmetic

SMT

A short overview of theories, models, decision procedures

Architecture of an SMT solver



What is a theory in mathematical logic?

- When we talk about **well-formed formulas with non-binary variables**, we have to say exactly what type of formulas are allowed
- and, what it means for assignments to *satisfy such formulas*
- This brings us to the notions **theory** and **models** in mathematical logic

Building up a theory

- First we define the syntax for writing formulas
 - A *signature* $\Sigma = (\Sigma_F, \Sigma_P, V)$
 - set of *function symbols* Σ_F , e. g., $\{+, -, f, g, \sin, \dots\}$
 - set of *predicate symbols* Σ_P
 - *arity* of each function: *arity*: $\Sigma_F \rightarrow \mathbb{N}$
 - 0 *arity* functions are constants
 - V : set of *variables*
 - *Terms*(Σ, V)
 - Elements of V are terms
 - If $t_1, \dots, t_k \in \text{Terms}(\Sigma, V)$ and $f \in \Sigma_F$ with arity k , then $f(t_1, \dots, t_k) \in \text{Terms}(\Sigma, V)$
 - *Ground terms* are terms without variables
- $\Sigma_F = \{0, +\}, \Sigma_P = \{<\}$
 - $\text{arity}(0) = 0$
 - $\text{arity}(+) = 2$
 - $\text{arity}(<) = 2$
 - $V = \{x, y, z\}$
 - Terms defined by this signature are $x, y, z, +(x, y), ++(x, y), 0, \dots$

Terms to Formulas

- **Atomic formulas** AF
 - True, False
 - If $t_1, \dots, t_k \in Terms(\Sigma, V)$ and $p \in \Sigma_P$ with arity k , then $p(t_1, \dots, t_k) \in AF(\Sigma, V)$
 - A *literal* is an AF or its negation
 - Set of all atomic formulas $AF(\Sigma, V)$
 - **Quantifier free formulas** $QFF(\Sigma, V)$
 - AF
 - if $\phi_1, \phi_2 \in QFF$ then
 - $\neg\phi_1 \in QFF, \phi_1 \wedge \phi_2 \in QFF, \phi_1 \vee \phi_2 \in QFF, \phi_1 \rightarrow \phi_2 \in QFF$
 - Set of all quantifier free formulas $QFF(\Sigma, V)$
 - **First order formulas** is the set of quantifier free formulas under universal and existential quantifiers
 - **Bound variables** are those that are attached to quantifiers
 - **Free variables**: variables not bound
 - **Sentence**: First order formula with no free variables
 - **Theory** (Σ, V) set of all sentences over (Σ, V)
- $x < y$
 - $+(x, y) = +(y, x)$
 - $+(x, y) = 0 \wedge x > y$
 - $\forall x, \exists y: +(x, y) = 0$
 - $\forall x, \exists y: x < y$
 - $\forall x, \exists y: +(x, y) = x$
 - $\exists x: +(x, c) = x$

Models for theories

This notion of model from mathematical logic is not to be confused with the notion of a model for a computational or physical process

- A *model* gives meanings or *interpretations* to formulas in theory T
- A model M for $T = \text{Theory}(\Sigma, V)$ has to define
 - A *domain* $|M|$
 - interpretations of all functions and predicate symbols
 - $M(f): |M|^n \rightarrow |M|$ if $\text{arity}(f) = n$
 - $M(p) \subseteq |M|^n$ if $\text{arity}(p) = n$
 - Assignment $M(x) \in |M|$ for every variable $x \in V$
- A *formula* ϕ is *true in* M if it evaluates to true under the given interpretations over domain M

Example model for $\Sigma = \{0, +, <\}$

$$|M| = \{a, b, c\}$$

$$M(0) = a$$

$$M(<) = \{\langle a, b \rangle, \langle a, c \rangle, \langle b, c \rangle\}$$

$M(+)$	a	b	b
a	a	b	c
b	b	c	a
c	c	a	b

□

$$\text{if } M(x) = a, M(y) = b$$

$$\text{then } M(+ (x, y)) \text{ is } M(+)(M(x), M(y)) = M(+)(a, b) = b$$

$$M(+ (+ (x, y), y)) = c$$

$$M \models \forall x \exists y + (x, y) = x$$

We say that the model M *T-satisfies* the formula ϕ

Decision procedures

Given a theory T a **theory solver or a decision procedure** for T takes as input a set of **literals** ϕ (atomic propositions) and determines whether ϕ is **T -satisfiable**, that is,

$\exists$ a model M such that $M \models \phi$?

$\langle \backslash \textit{logic} \rangle$

A short overview of theories and models in mathematical logic

Example theories

- Uninterpreted functions (UF) $\Sigma_F := \{f, g, \dots\}$, $\Sigma_P := \{=\}$, $V := \{x_i\}$
 - $x_1 = x_2 \wedge x_3 \neq x_2 \wedge f(x_3) \neq f(x_2)$
- Difference logic $\Sigma_F := \{1, 2, \dots, -\}$, $\Sigma_P := \{<, \leq, =, >, \geq\}$
 - $x_1 - x_2 \geq k$, where $\geq \in \{<, \leq, =, >, \geq\}$
- Linear arithmetic
 - $4x - 3y + 6z \leq 10$
- Real arithmetic (nonlinear)
 - $4x^2 + 6y - 9z^3 \leq 5$
- Bit vectors
- Arrays
 - $x'[i] = x[i] + 1$

Example decision procedure 1: Difference logic

$$\phi = (x - y = 5) \wedge (z - y \geq 2) \wedge (z - x > 2) \wedge (w - x = 2) \wedge (z - w < 0)$$

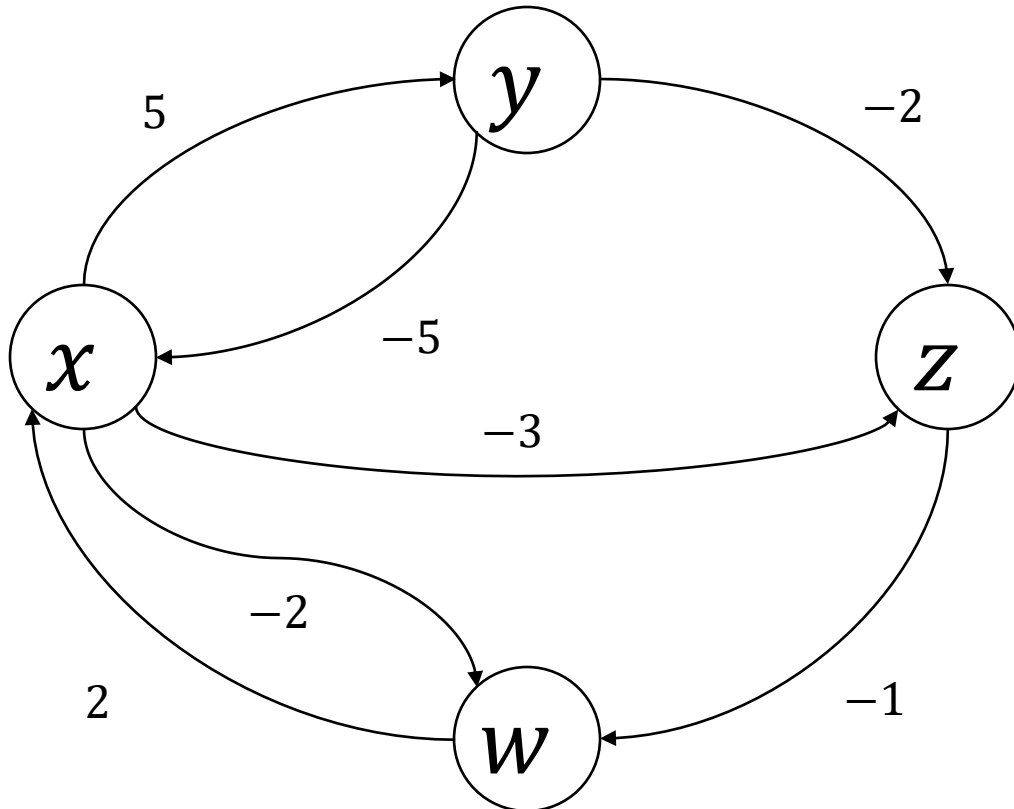
Decision procedure:

Convert each literal (AF) to $x_1 - x_2 \leq c$ form:

$$\begin{aligned} \phi' = & (x - y \leq 5) \wedge (y - x \leq -5) \\ & \wedge (y - z \leq -2) \wedge \\ & (x - z \leq -3) \wedge \\ & (w - x \leq 2) \wedge (x - w \leq -2) \\ & (z - w \leq -1) \end{aligned}$$

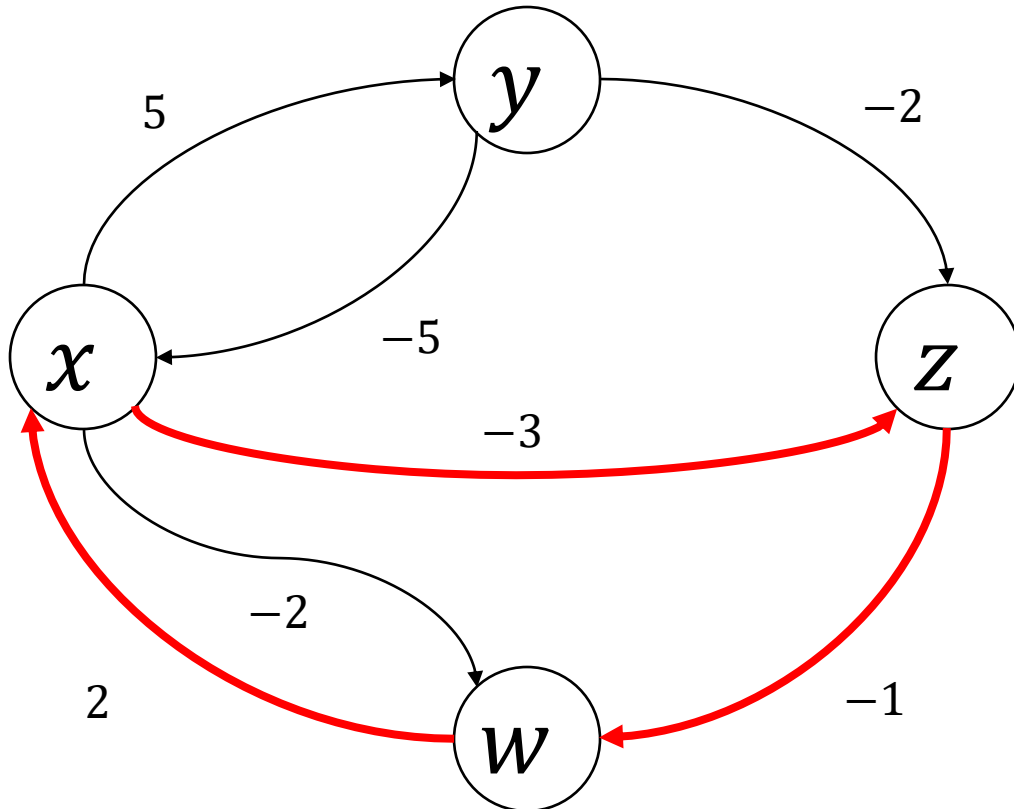
$$\begin{aligned}\phi' = & (x - y \leq 5) \wedge (y - x \leq -5) \\ & \wedge (y - z \leq -2) \wedge \\ & (x - z \leq -3) \wedge \\ & (w - x \leq 2) \wedge (x - w \leq -2) \\ & (z - w \leq -1)\end{aligned}$$

Construct a graph with edge from $x \rightarrow^c y$ for each literal ϕ'



$$\begin{aligned}\phi' = & (x - y \leq 5) \wedge (y - x \leq -5) \\ & \wedge (y - z \leq -2) \wedge \\ & (x - z \leq -3) \wedge \\ & (w - x \leq 2) \wedge (x - w \leq -2) \\ & (z - w \leq -1)\end{aligned}$$

Construct a graph $G_{\phi'}$, with edge from $x \rightarrow^c y$ for each literal ϕ'



Proposition. ϕ is satisfiable iff $G_{\phi'}$ is negative cycle free.

Proof: Exercise.

Example decision procedure 2: Uninterpreted functions (UF)

$$\phi = x_1 = x_2 \wedge (x_2 = x_3) \wedge (x_4 = x_5) \wedge (x_5 \neq x_1) \wedge (F(x_1) \neq F(x_3))$$

Decision procedure

1. Put all variables and function instances in their own classes
2. If $t_1 = t_2$ is a literal then merge the classes containing them; do this repeatedly
3. If t_1 and t_2 are terms in the same class then merge classes containing $F(t_1)$ and $F(t_2)$; repeat
4. If $t_1 \neq t_2$ is a literal in ϕ and they belong to the same class then return unsat else return sat t_1 and t_2

Example decision procedure 2: Uninterpreted functions (UF)

Initial classes $\phi = x_1 = x_2 \wedge (x_2 = x_3) \wedge (x_4 = x_5) \wedge (x_5 \neq x_1) \wedge (F(x_1) \neq F(x_3))$

Classes $\{x_1\} \{x_2\} \{x_3\} \{x_4\} \{x_5\} \{F(x_1)\} \{F(x_3)\}$

$\{x_1, x_2, x_3\} \{x_4, x_5\} \{F(x_1)\} \{F(x_3)\}$

$\{x_1, x_2, x_3\} \{x_4, x_5\} \{F(x_1), F(x_3)\}$

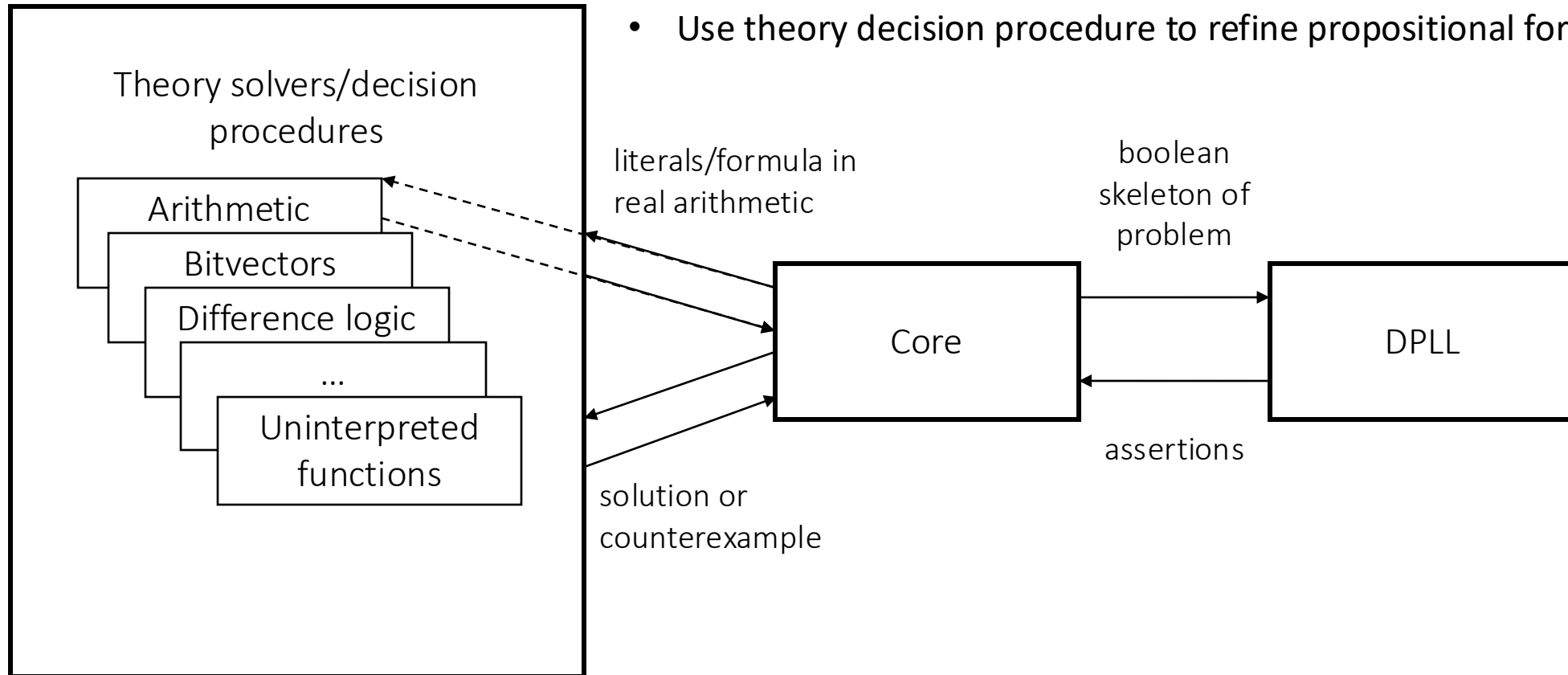
Unsat

Return to SMT

$$\phi \equiv g(a) = c \wedge f(g(a)) \neq f(c) \vee g(a) = d \wedge c \neq d$$

Several approaches, lazy approach:

- Abstract ϕ to propositional form
- Feed to DPLL
- Use theory decision procedure to refine propositional formula a guide SAT



DPLL for SMT

Input: A formula F in CNF form over theory T

Output: $I \models F$ or UNSAT

Let F^B be the abstraction of F

while true do

if DPLL(F^B) is unsat **then return** UNSAT

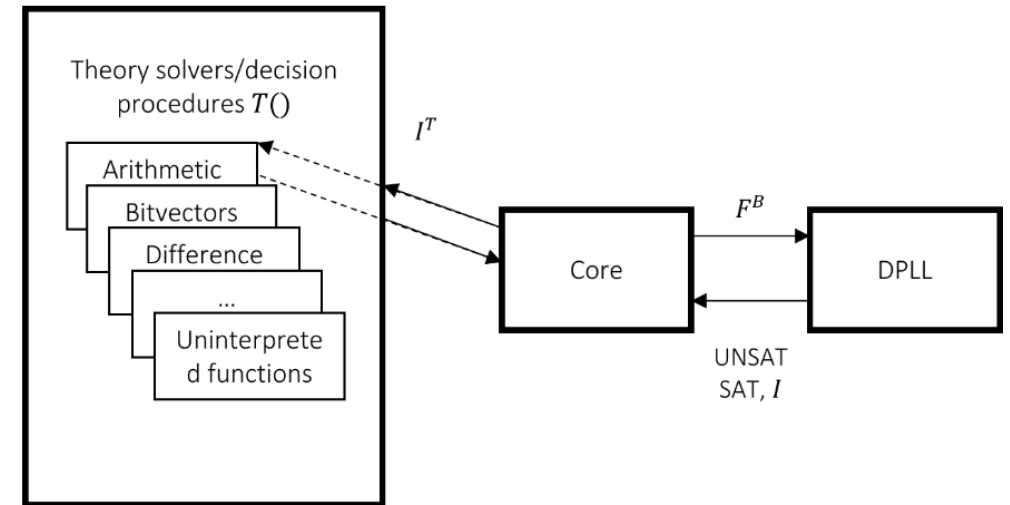
else

 Let I be the model returned by DPLL

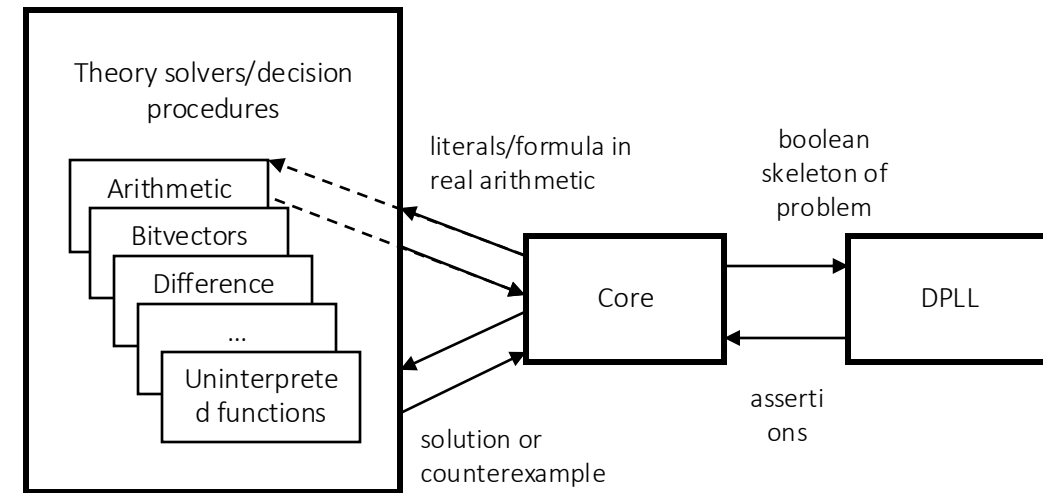
 Assume I is represented as a formula

if $T(I^T)$ is sat **then return** SAT and the model returned by $T()$

else $F^B := F^B \wedge \neg I$



$$\phi \equiv \underbrace{g(a) = c}_1 \wedge \underbrace{f(g(a)) \neq f(c)}_{\bar{2}} \vee \underbrace{g(a) = d}_3 \wedge \underbrace{c \neq d}_{\bar{4}}$$



- send $\{1, \bar{2} \vee 3, \bar{4}\}$ to DPLL
- returns model $\{1, \bar{2}, \bar{4}\}$
- UF solver concretizes to $g(a) = c, f(g(a)) \neq f(c), c \neq d$
- checks this as UNSAT
- send $\{1, \bar{2} \vee 3, \bar{4}, \bar{1} \vee 2 \vee 4\}$ to DPLL
- returns model $\{1, 2, 3, \bar{4}\}$
- UF solver concretizes and finds this to be UNSAT
- send $\{\bar{1}, \bar{2} \vee 3, \bar{4}, \bar{1} \vee 2 \vee 4, \bar{1} \vee \bar{2} \vee \bar{3} \vee 4\}$ to DPLL
- returns UNSAT

Assignments

- HW1
- Learn z3
 - <https://ericpony.github.io/z3py-tutorial/guide-examples.htm>

Readings

- Chapter 7.5.3 of CPSBook on using SAT/SMT for verification
- Read chapter 4 for next week
- Reading more about decision procedures

